AlphaEvolve on an 8-bit level-set advection kernel

Deforming bubble, WENO5 advection, RAPTOR fp8 truncation, and an interface-weighted score.

1 Question

Can the WENO5 advection kernel run in 8-bit (RAPTOR-emulated) while the level set stays within a tolerable error of an FP64 shadow run, and can AlphaEvolve improve the low-precision seed?

2 Setup

App: Deforming Bubble mini-app (Flash-X level set), 64x64, tmax=2, 1 rank. Kernel: `advectWeno` in `app/src/advect.cpp`. Raptorize added modes fp64, fp8e4m3, fp8e5m2, lowprec (native float seed), plus an FP64 shadow run and error columns in `history.csv`.

3 AlphaEvolve experiment

Parameter	Value
Experiment	exp-invaluable-boar
Model	gemini-3-flash-preview
Max programs	20
Concurrency	4
Timeout per eval	600 s
Eval run	nx=64 ny=64 tmax=2 plot_interval=0 precision=lowprec
Score	-log10(RMS over rows of err_band)
Seed score	8.318818 (5113 rows, rms err_band 4.799e-9)
Best score	8.576261 (rms err_band 2.653e-9)
Programs evaluated	19 (18 succeeded, 1 failed)

Evaluation: build with `make -j4`, audit the EVOLVE-BLOCK (no double, no `#include`, `*LowPrec` names, no fp64 asm), run the lowprec case, read `err_band` from `history.csv`, score `-log10(rms)`. Rules violations score null.

4 Timing

Run	Steps	Wall time (s)
fp64 animation	5112	341.5
fp8e4m3 original animation	5112	341.5
fp8e4m3 optimized animation	5112	337.7
AlphaEvolve job	19 programs	~20 min

5 Results at t=2

Mode	area	err_rms	err_band
fp64	0.1772	0	0
fp8e4m3 (original)	0.2229	3.216e-2	1.972e-2
fp8e4m3 (optimized seed)	0.2229	3.216e-2	1.972e-2
lowprec seed (before)	–	–	4.799e-9 (eval)
lowprec seed (after)	–	–	2.653e-9 (eval)

6 Level-set behavior

The fp64 run deforms and splits the bubble by t=2. The fp8e4m3 run follows the same topology but develops ragged contours and spurious phi=0 islands along the thin filament; err_band rises from ~1.3e-2 at t=0.5 to ~2.0e-2 at t=2, and wrong-sign cells reach ~16%.

The optimized seed improves the native float path (err_band 1.8x lower in the eval run) but does not change the fp8 path, because RAPTOR truncates the FP64 `advectWeno`, not the evolved float seed. The fp8 animation is bit-identical before and after optimization.

7 Qualitative and quantitative

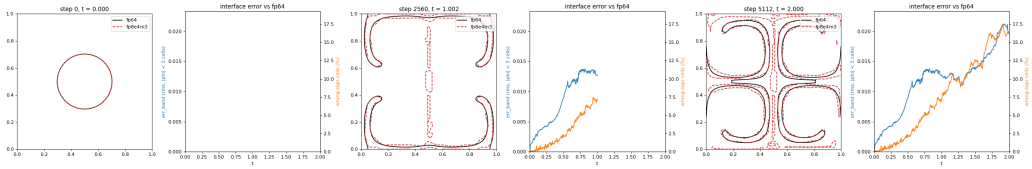


Figure 1: fp64 (solid black) vs. fp8e4m3 (dashed red) $\phi = 0$ contours, with err_band and wrong-sign cells vs. time beside them. Frames at $t \approx 0, 1, 2$.

8 Conclusion

AlphaEvolve improved the float seed's interface error, but the 8-bit (RAPTOR) path is unchanged. To affect fp8 output, the optimization target must be the RAPTOR-truncated FP64 function itself.